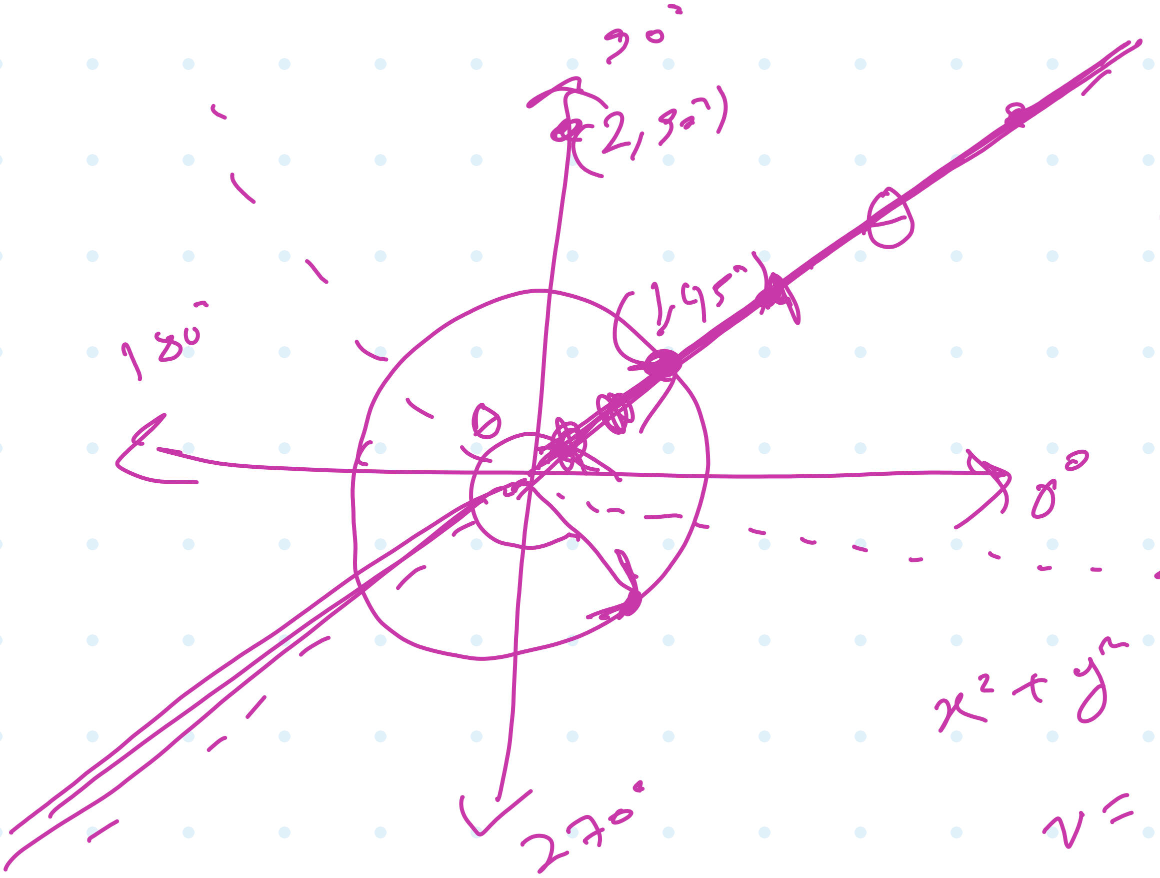


(r, θ)

$x^2 + y^2 = 1$

$r = 1$

$x = y$

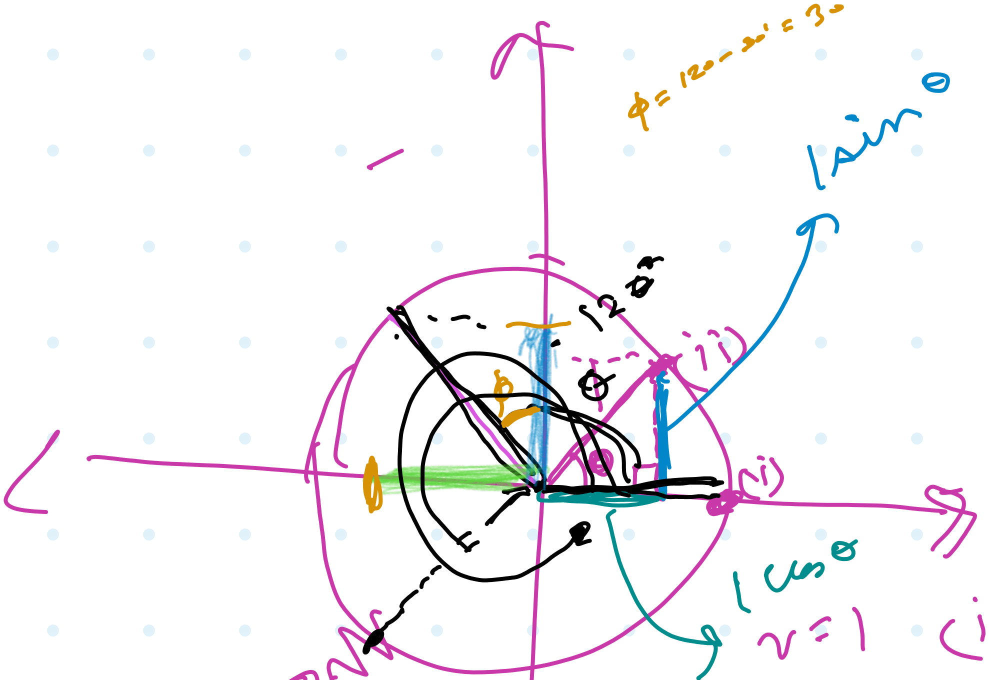
$x - y = 0$

$\theta = 45°$

$\theta = 45°$

$x - y = 0$

$x \geq 0$



$r = 1$

	θ	sin θ	cos θ
(i):	0°	0	1
(ii)	45°	$1/\sqrt{2}$	$1/\sqrt{2}$
(iii)	120°	$\sqrt{3}/2$	$-1/2$

Domain Range Codomain

sin x sine

(domain) x $\sin(\ )$ $\longrightarrow$ y (range)

R^1 R^1 [2, 3) 0, $\frac{22}{7}$, $\frac{\pi}{4}$, $\sqrt{-59}$, $6a^2$, $2i$

$$f(x) = \frac{1}{x}$$

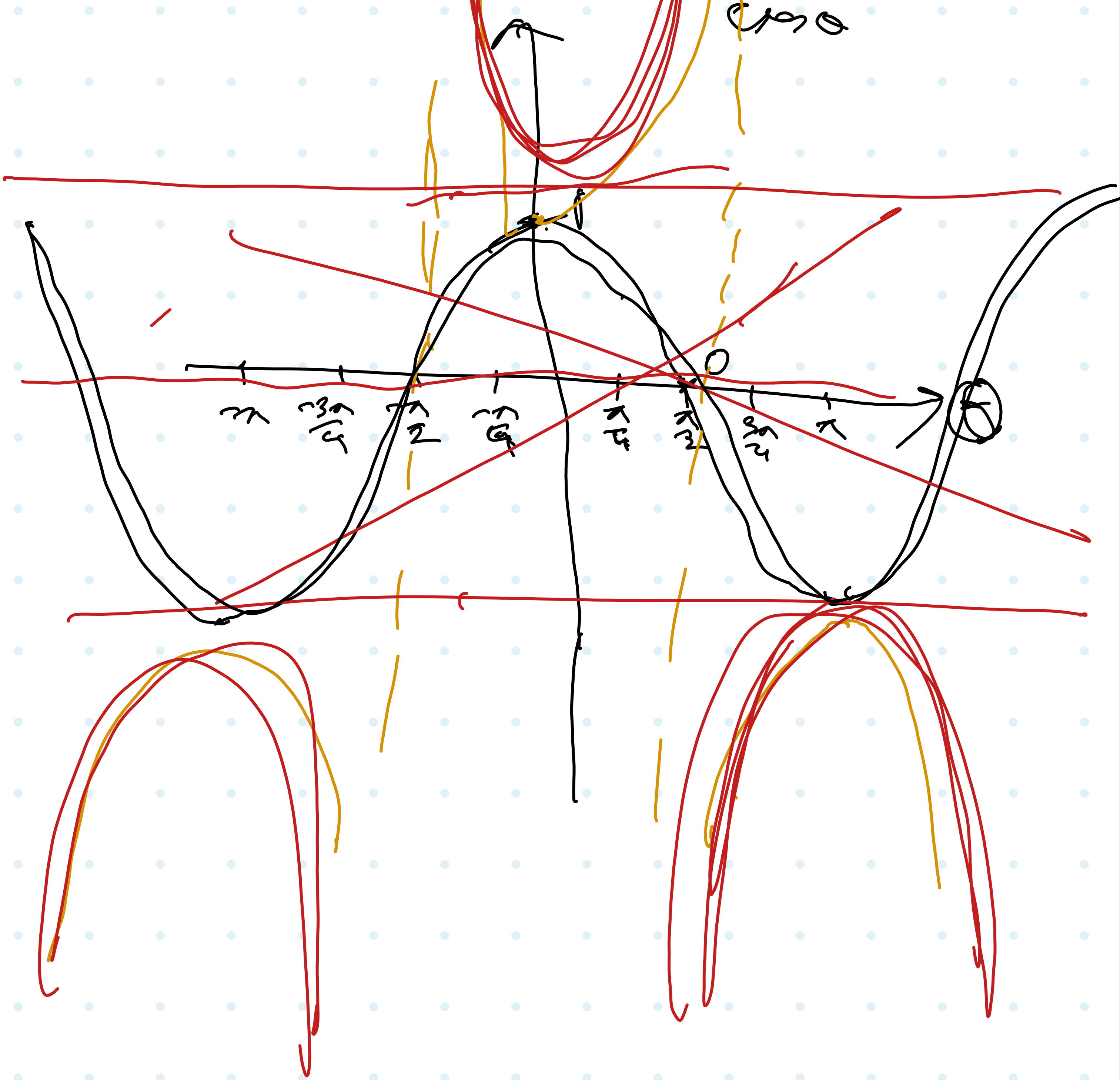
D: $R^1 - \{0\}$ $\left\{ Q, \bar{Q} \right\}$

$$f(x) = \frac{1}{(x-2)^2}$$ D: $R^1 - \{2\}$

$\sin 270^\circ =$ $\sin\theta \in [-1, 1]$
$\sin(-90 + 360)$
$= \sin(-90) = -\sin 90$
$= -1$

$$\text{range}(\sec(\theta)) = \boxed{(-\infty, -1] \cup [1, \infty)}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

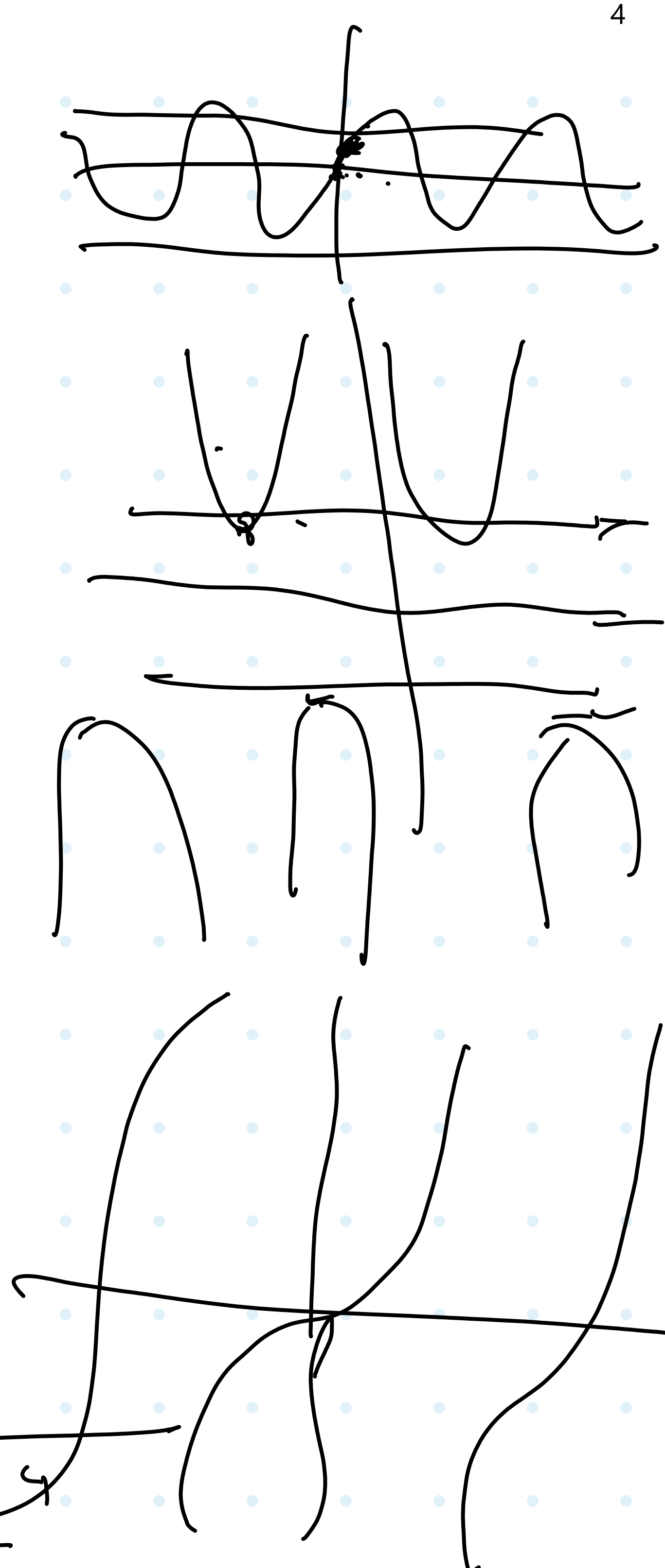


$\sin x$

$\cos x$

$\sec x$

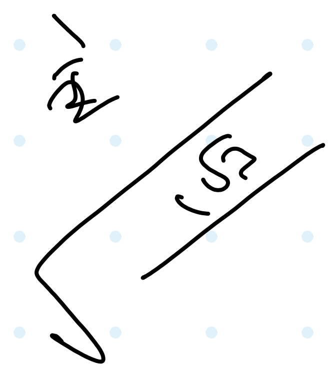
$\tan x$



$\tan 15$:

$\tan 30 =$

$$\tan 30 = \frac{2 \tan 15}{1 - \tan^2 15}$$



$$\frac{2y}{1-y^2} = \frac{1}{\sqrt{3}}$$

$$2\sqrt{3}y = 1 - y^2$$

$$y^2 + 2\sqrt{3}y - 1 = 0$$

$$y = \frac{-2\sqrt{3} \pm \sqrt{12+4}}{2}$$

$$\sqrt{\left(\frac{1}{2} + \frac{\sqrt{3}}{4}\right)} \qquad y = \frac{-2\sqrt{3} + 4}{2} = \boxed{-\sqrt{3} + 2}$$

$$\frac{\sqrt{3}}{2} = 2c^2 - 1 \qquad \sin 75 =$$

$$\frac{\sqrt{3}}{2} + 1 = 2c^2$$

$$s\,75 = c\,15$$

$$c\,30 = 2(c\,15)^2 - 1$$

$\boxed{3\text{-}3}$ $\boxed{9.}$

$$\cos\left(\frac{3\pi}{2} + x\right)$$

a

$$= \cos\left(2\pi - \underbrace{\left(\frac{\pi}{2} - x\right)}_{a}\right)$$

$$= \cos(a + 2\pi)$$

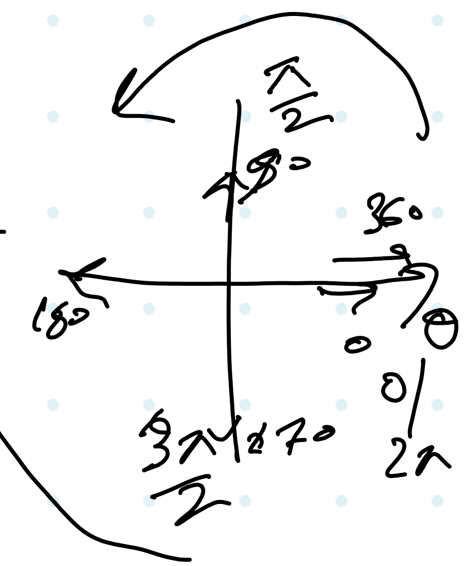
$$= \cos(a)$$

$$= \cos\left(-\left(\frac{\pi}{2} - x\right)\right)$$

$$= \cos\left(\frac{\pi}{2} - x\right)$$

$$\neq \sin(x)$$

θ



$$\cos\left(\frac{3\pi}{2} + x\right)$$

$$= \cos\left(\pi + \left(\frac{\pi}{2} + x\right)\right)$$

$$= -\cos\left(\left(\frac{\pi}{2} + \cancel{2x}\right)\right)$$

$$= -\cos\left(\frac{\pi}{2} + \cancel{2x}\right)$$

$$= \sin x$$

$\cos\theta =$

$\cos(2\pi + \theta)$

$= \cos(\theta + 2\pi)$

$= \cos(\theta - 2\pi)$

$= \cos(\theta + 4\pi)$

A B C D

$\sin x \quad \cos x \quad \tan x \quad \cot(x)$

$A\,B\,(C + D)$

$S.C.\left(\frac{S}{C} + \frac{C}{S}\right) = \cancel{S.C}\left(\frac{S^2 + C^2}{\cancel{SC}}\right)$

$= 1$ Q.E.D.

$\cos\left(\boxed{\frac{3\pi}{2}} - x\right) \qquad \sin\left(\frac{3\pi}{2} - x\right)$

$\cos\left(2\pi - \left(\frac{\pi}{2} + x\right)\right) \qquad S(\;\text{—}\;)$

$\cos\left(-\left(\frac{\pi}{2} + x\right)\right) \qquad S(\;\text{—}\;)$

$\cos\left(\frac{\pi}{2} + x\right) \qquad -S\left(\frac{\pi}{2} + x\right)$

$\cos\left(\frac{\pi}{2} - (-x)\right) \qquad -S\left(\frac{\pi}{2} - (-x)\right)$

$-C(-x)$

$\sin(-x)$

$-\sin x$

$C_N = -\sin x$

$C_D = -\cos x$

$C = \frac{C_N}{C_D} = \tan x$

$$\cos\left(\frac{3\pi}{2}+x\right)$$

$$= \cos\left(\frac{3\pi}{2}\right)\cos x - \sin\left(\frac{3\pi}{2}\right)\sin(x)$$

$$= \sin x$$

$\boxed{11.}$ $\cos\left(\frac{3\pi}{4}+x\right) - \cos\left(\frac{3\pi}{4}-x\right)$

A B

$$= -2\sin\left(\frac{3\pi}{4}\right)\sin(x)$$

$$= -\sqrt{2}\sin x$$

$\boxed{15}$ $$\frac{\cot 4x}{\cot x} = \frac{s5x - s3x}{s5x + s3x}$$

$$= \frac{\cancel{2}\, c4x\, s\, x}{\cancel{2}\, s4x\, c\, x}$$

$$= \frac{\cot 4x}{\cot x}$$

$\boxed{23}$. $$\frac{4\tan x(1-\tan^2 x)}{1-6\tan^2 x+\tan^4 x}$$

$$T4 = \frac{2T2}{1-(T2)^2}$$

$$T_2 = \frac{2T_1}{1-T_1^2}$$

$$\frac{2\left(\frac{2T_1}{1-T_1^2}\right)}{1-\left(\frac{2T_1}{1-T_1^2}\right)^2}$$

$$\frac{2\cdot 2T_1\left(1-T_1^2\right)}{\left(1-T_1^2\right)^2-\left(2T_1\right)^2}$$

$$\frac{4T_1\left(1-T_1^2\right)}{T_1^4-6T_1^2+1}$$

$$c_3 = 4c^3 - 3c$$

$$c_6 = 2c_3^2 - 1$$

$$= 2(4c_1^3 - 3c_1)^2 - 1$$

$$= 2(16c_1^6 - 24c_1^4 + 9c_1^2) - 1$$

$$= \underline{32c_1^6 - 48c_1^4 + 18c_1^2 - 1}$$